

iMHEA Data Processor — User Manual

Version 1.0 • July 2026

Table of Contents

1. Introduction

The iMHEA Data Processor converts raw sensor records from headwater catchment monitoring — tipping-bucket rain gauge events and water-level or discharge logger files — into quality-controlled rainfall and streamflow series at high, hourly, and daily resolution, and computes 59 hydrological and 13 climate indices describing water yield, regulation, flashiness, frequency, duration, timing, and rainfall intensity.

The methods implement Ochoa-Tocachi et al. (2018), *High-resolution hydrometeorological data from a network of headwater catchments in the tropical Andes*, Scientific Data 5:180080. The software is a validated Python translation of the original iMHEA MATLAB scripts: streamflow and baseflow outputs reproduce the published dataset exactly, and every known defect of the original code has been fixed (a compatibility mode can reproduce the legacy behaviour; see section 11).

Citing. If you use results from this software in a publication, cite the paper above.

2. Preparing your data

2.1 Raw file format

The software reads the standard iMHEA raw CSV layout. One file per sensor:

Rain gauge files (tipping-bucket events):

```
Date,Event mm,Flag
24/05/2001 00:00:01,0,I
24/05/2001 00:41:23,0.2,
03/04/2004 00:00:01,,V
```

Water level / discharge files:

```
Date,Level cm,Flow l/s,Flag
17/07/2012 14:00:00,8.1000,2.5582,"D,P"
```

Rules:

- Dates in dd/mm/yyyy HH:MM:SS, sorted ascending.

- Decimal point (.), comma-separated columns.
- Flags: I = logger launched, D = data downloaded, X = erroneous tip removed, P = anomalous intensity, V = data gap. A V row with an **empty value** marks the start of a gap — this is how the software knows data are missing rather than dry/zero.
- Encoding UTF-8 (a byte-order mark is tolerated); line endings may be Windows, Unix, or legacy Mac.
- Rainfall values are millimetres per tip; level in centimetres; flow in litres per second.

A machine-readable version of this specification is included as docs/DATA_FORMAT.md.

2.2 What you need to know about each catchment

Parameter	Meaning	Typical iMHEA values
Area	catchment area draining to the weir [km ²]	0.5 – 16
Bucket	rain gauge resolution [mm per tip]	0.1, 0.2, or 0.254
Weir geometry	V-notch height <i>a</i> [m], rectangular width <i>b</i> [m]	0.30 / site-specific

3. Getting started

On launch, the start screen offers three paths:

- **Process my data** — creates an empty project to which you add your own catchments and files.
- **Explore the iMHEA network** — asks for the folder containing iMHEA_raw/ and loads the complete 2018 network configuration (28 stations, 12 pairs) ready to run.
- **Open project...** — reopens a saved .imhea project file.

A **project** stores your catchment configurations, file locations, and options. Save it from the toolbar (*Save project*); results are recomputed on demand and are not stored in the project file.

The toolbar also holds **Run pipeline**, **Export**, and the **EN/ES** language switch.

4. Configuring a catchment

Select a catchment in the sidebar and open its **Setup** tab.

1. Enter the code (e.g. MYC_01), area, and bucket size.
2. Under **Discharge**, add the level/flow file(s). If the file has a flow column it is used directly; choose *Level → rating curve...* to convert water level instead (section 4.1). Rain-only stations simply leave discharge empty.

3. Under **Rain gauges**, add one file per gauge. Multiple gauges are cross-checked and averaged automatically (section 5).
4. Each file shows a summary line — row count, time step, detected columns, period. Check these before running.

4.1 Rating curve editor

For stations recording water level only, the discharge is computed with the compound sharp-crested weir equation (90° V-notch inside a rectangular section):

- Within the notch ($h \leq a$): $Q = C1 \cdot h^{e1}$
- Above it: $Q = C1 \cdot (h^{e1} - (h-a)^{e1}) + C2 \cdot b \cdot (h-a)^{e2}$

The editor lets you set a , b and the four coefficients (defaults: $C1 = 1.37$, $e1 = 2.5$, $C2 = 1.77$, $e2 = 1.5$) and shows the resulting curve live. Check b against your weir drawings — it is site-specific.

5. Running the pipeline

Press **Run pipeline** with a catchment selected. Processing runs in the background; messages stream into the **Log** tab. What happens, in order:

1. **Depuration** — logger bounce tips arriving ≤ 1.1 s apart are removed.
2. **Grid selection** — the working resolution is the median time step of the discharge record (5 min typical; rain-only stations use 5 min).
3. **Rainfall disaggregation** — tip events become a continuous intensity series via cubic-spline interpolation of the cumulative curve (Sadler & Busscher 1989; Wang et al. 2008), preserving total volume; storms are separated at 0.2 mm/h, intensities capped at 127 mm/h, single tips spread at 3 mm/h.
4. **Gauge cross-filling** — every pair of gauges is compared by double-mass analysis; gaps are filled proportionally only when the correlation $R \geq 0.99$. Catchment rainfall is the average of the filled series.
5. **Discharge averaging** to the grid; normalisation by area.
6. **Aggregation** to hourly and daily products; **baseflow separation** (UKIH smoothed-minima and Chapman two-parameter filter).
7. **Indices** — 59 hydrological + 13 climate indices (section 8).

The status bar reports the runtime (a few seconds per catchment typically).

6. Results

The **Results** tab shows summary cards (period, runoff ratio, baseflow index, gap percentage) and one of six figures, at daily, hourly or high resolution:

Figure	Shows
--------	-------

Hydrograph	inverted rainfall bars over the flow series with shaded baseflow
Flow duration curve	percentage of time each flow is exceeded (log scale)
Intensity-duration curve	maximum rainfall intensity vs duration, 5 min – 2 days
Monthly regime	precipitation and runoff climatology, mm/month
Data coverage	timeline of valid data and gaps per variable
Double-mass curve	cumulative rainfall vs cumulative runoff; breaks in slope reveal changes in behaviour or instrumentation

Figures are interactive (zoom, pan) and can be saved as PNG/PDF/SVG from the toolbar above the plot.

7. Gap-filling report

The **Gap-filling report** button (Results tab) lists every double-mass comparison of the last run: gauge pair, whether filling was applied, the correlation R, the slope M, and how many intervals were filled in each series. Use it to document quality control and to detect gauges that drift apart (R below 0.99 means no filling was applied — investigate why).

8. The indices

The **Indices** tab lists all values; *Copy table* and *Save CSV...* export them. Indices follow Olden & Poff (2003) nomenclature where applicable. Qxx denotes the flow exceeded xx% of the time; percentiles use Gringorten plotting positions. All flow indices are computed from daily mean flow in l/s/km²; rainfall intensity indices from the 5-minute grid.

8.1 Hydrological indices (59)

#	Code	Description	Units
1	QDMIN	Minimum daily flow	l/s/km ²
2	Q95	Flow exceeded 95% of the time (low flow)	l/s/km ²
3	DAYQ0	Days with zero flow per year	day
4	PQ0	Proportion of days with zero flow	–
5	QMDRY	Mean daily flow of	l/s/km ²

			the driest month	
6	QDMAX	Maximum daily flow	1/s/km ²	
7	Q10	Flow exceeded 10% of the time (high flow)	1/s/km ²	
8	QDMY	Annual mean daily flow	1/s/km ²	
9	QDML	Long-term mean daily flow	1/s/km ²	
10	Q50	Median flow from the FDC	1/s/km ²	
11	BFI1	Baseflow index, UKIH smoothed-minima method	–	
12	K1	Recession constant, UKIH method	–	
13	BFI2	Baseflow index, Chapman 2-parameter filter	–	
14	K2	Recession constant, Chapman filter	–	
15	RANGE	Discharge range QDMAX/QDMIN	–	
16	R2FDC	Slope of the FDC between 33% and 66% exceedance	–	
17	IRH	Hydrological regulation index	–	
18	RBI1	Richards-Baker flashiness index (annual)	–	
19	RBI2	Richards-Baker flashiness index (seasonal)	–	
20	DRYQMEAN	Driest-month flow / mean monthly flow	–	
21	DRYQWET	Driest-month flow / wettest-month flow	–	
22	SINDQ	Streamflow seasonality index	–	
23	QYEAR	Average annual	mm	

			discharge	
24	RRa	Annual runoff ratio	–	
		QYEAR/PYEAR		
25	RRm	Monthly runoff ratio	–	
26	RRl	Long-term runoff ratio	–	
27	MA5	Skewness of daily flows: mean/median	–	
28	MA41	Mean annual runoff	1/s/km ²	
29	MA3	Coefficient of variation of daily flows	–	
30	MA11	(Q25 – Q75) / median	–	
31	ML17	7-day minimum flow / mean annual flow	–	
32	ML21	CV of 30-day minimum flows	–	
33	ML18	CV of 7-day minimum flows	–	
34	MH16	High flow: Q10 / median	–	
35	MH14	Median 30-day maximum flow / median	–	
36	MH22	Mean high-flow volume above 3× median / median	day	
37	MH27	Mean peak above Q25 / median	–	
38	FL3	Low pulses below 5% of mean daily flow	1/yr	
39	FL2	CV of annual low-pulse counts (FL1)	–	
40	FL1	Low pulses below Q75	1/yr	
41	FH3	High pulses above 3× median daily flow	1/yr	

42	FH6	High-flow events above $3\times$ median monthly flow	1/yr
43	FH7	High-flow events above $7\times$ median monthly flow	1/yr
44	FH2	CV of annual high-pulse counts (FH1)	–
45	FH1	High pulses above Q25	1/yr
46	DL17	CV of low-pulse durations (DL16)	–
47	DL16	Mean low-pulse duration below Q75	day
48	DL13	Mean 30-day minimum flow / median	–
49	DH13	Mean 30-day maximum flow / median	–
50	DH16	CV of high-pulse durations (DH15)	–
51	DH20	Mean high-pulse duration above median/0.75	day
52	DH15	Mean high-pulse duration above Q25	day
53	TH3	Longest period with no flood (above Q10), fraction of year	–
54	TL2	CV of TL1	–
55	TL1	Median Julian day of the annual 1-day minimum	–
56	RA8	Rate of flow reversals between days	1/day
57	RA5	Fraction of days with rising flow	–
58	RA6	Median rise rate, log	–

59	RA7	space Median fall rate, log space	–
----	-----	---	---

8.2 Climate indices (13)

#	Code	Description	Units
1	PYEAR	Average annual precipitation	mm
2	DAYPO	Days with zero precipitation per year	day
3	PPO	Proportion of days with zero precipitation	–
4	PMDRY	Precipitation of the driest month	mm
5	SINDX	Seasonality index (0 none – 1 extreme)	–
6	PVAR	Coefficient of variation of daily precipitation	–
7	RMED1D	Median annual maximum 1-day precipitation	mm
8	RMED2D	Median annual maximum 2-day precipitation	mm
9	RMED1H	Median annual maximum 1-hour precipitation	mm
10	iMAX1D	Maximum 1-day intensity	mm/h
11	iMAX2D	Maximum 2-day intensity	mm/h
12	iMAX1H	Maximum 1-hour intensity	mm/h
13	iMAX15M	Maximum 15-minute intensity	mm/h

(The published dataset lists two further rows, ETYEAR and RPPE, derived from external evapotranspiration data; they are not computed by this software.)

9. Paired catchments

The iMHEA design compares paired catchments (e.g. conserved vs intervened). Create a pair with **+ Add pair** after configuring both catchments. Running a pair:

- resamples both to the coarser common resolution;
- cross-fills **rainfall** between the catchments by double-mass analysis (discharge is never filled);
- produces combined high-resolution/hourly/daily tables and side-by-side indices, with comparison figures (overlaid hydrographs, FDCs, IDCs).

10. Exporting

Export writes, to a folder you choose:

- iMHEA_<CODE>_HRes_processed.csv — high-resolution [Date, Rainfall, Flow]
- iMHEA_<CODE>_1hr_processed.csv — hourly
- iMHEA_<CODE>_1day_processed.csv — daily, including baseflow
- for pairs, the corresponding _01/_02 files in the published iMHEA format (dates dd/mm/yyyy HH:MM:SS, fixed-width values, NaN for gaps).

Indices export from the Indices tab (CSV or clipboard).

11. MATLAB compatibility mode

During translation, nine defects were found in the original MATLAB code (documented in docs/CODE_REVIEW.md and quantified in docs/VALIDATION.md). By default this software computes the **corrected** values. Ticking *MATLAB-compatible mode* (Setup tab) reproduces the legacy behaviour where feasible — useful when comparing against the 2018 published dataset. The differences that matter most:

- **BFI1/K1**: published values are inflated for records with data gaps (gap blocks entered the baseflow line as turning points).
- **TL1/TL2**: published timing values are unreliable after the first year of each record (day-of-year indexing defect).
- **FH7**: published values are double-normalised (too small by roughly the number of years).

12. Command-line reference

For batch work, the imhea command (requires the Python package):

```
imhea process --code MYC_01 --area 2.63 --bucket 0.2 \  
  --flow flow.csv --gauge rg1.csv --gauge rg2.csv --out ./processed  
imhea process --code RAIN_01 --bucket 0.1 --gauge rg.csv --out ./out
```

```
imhea pair --hres1 A_HRes.csv --hres2 B_HRes.csv --site XYZ --out ./out
imhea network --data-root ./Scripts --out ./valout
imhea --version
```

--level-to-flow (with --weir A B and --coeff C1 E1 C2 E2) converts a level column through the rating curve; --matlab-compat enables the compatibility mode.

13. Frequently asked questions

How much data is needed for meaningful indices? At least one full year; the software computes indices for any record length without warning, so interpret short-record results cautiously (annual statistics treat partial years as full ones, as in the original methodology).

Why does my rainfall total change slightly after processing? Depuration removes physically impossible double tips (≤ 1.1 s apart); the log reports the removed volume. Disaggregation itself conserves volume.

Gaps appear where I expect data. Gaps propagate from V-flag marker rows in the raw files. Check the Data coverage figure and the raw files' flag column.

Can I process data from non-imHEA loggers? Yes — convert to the raw CSV layout of section 2.1. Only the date format, column layout, and the V gap convention are mandatory.

Where are my project's results stored? Results live in memory and are recomputed on demand; exported CSVs and the .imhea project file are the persistent artefacts.

14. References

- Ochoa-Tocachi, B.F., et al. (2018). High-resolution hydrometeorological data from a network of headwater catchments in the tropical Andes. *Scientific Data* 5, 180080.
- Gustard, A., Bullock, A., Dixon, J.M. (1992). *Low flow estimation in the United Kingdom*. Institute of Hydrology Report 108.
- Chapman, T. (1999). A comparison of algorithms for stream flow recession and baseflow separation. *Hydrological Processes* 13.
- Olden, J.D., Poff, N.L. (2003). Redundancy and the choice of hydrologic indices for characterizing streamflow regimes. *River Research and Applications* 19.
- Sadler, E.J., Busscher, W.J. (1989). High-intensity rainfall rate determination from tipping-bucket rain gauge data. *Agronomy Journal* 81.
- Wang, J., et al. (2008). A new treatment of depth-dependent errors in tipping-bucket rain gauge measurements. *J. Atmos. Oceanic Technol.* 25.